

LISP. Besides, LISP has tremendous applications in the Artificial Intelligence implementations of various robotics problems. Even today, learning LISP is considered a bonus for any robotics specialist.

Pascal, another early programming language, is often thought of as the industrial roboticist's bread and butter. This language was introduced mainly to inculcate good programming practices, but has stuck on as a powerful and useful language - in fact, it was one of the earliest languages to support pointers. The benefit of learning Pascal is that many Industrial Robot Languages are heavily Pascal based. Hence, even with their individual idiosyncrasies and differences, a knowledge of Pascal serves as a common base from which to learn.

Robotics programming languages - today

Robots as a 20th and 21st century phenomenon have been closely matched, and often surpassed by advances in the field of computer science. It is therefore unsurprising that the bulk of robotic 'code', so to speak, has manifested in established programming languages as and when they have been popular.

Perhaps the most popular language for robotics is C. C is a language that is close to the hardware, and hence is useful for those hardware libraries that are commonly used to direct robotic arms, actuators and motors. Further, C is well-suited to real-time performance of the robot. This has made C (and in modern times, C++), the language of choice for most robotics applications, even though it often takes longer in terms of time as well as code length to implement something in C as compared to Python or MATLAB. C's maturity and robustness have often lent it the moniker of 'the standard language' in robotics circles.

Perhaps the next most powerful languages are Python and



Source: Wikimedia

Simulation of the robot Robotis Darwin using a simulation software